1 **(a)** State Newton's three laws of motion.

First Law: .....

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Second Law: .....

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Third Law: .....

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..... [3]

(b) Using the appropriate laws of motion, answer the question for each given situation.

(i) A passenger claimed that he was sitting in the middle of a bus that was moving forward. The driver suddenly applied the brakes and a suitcase that was in the front of the bus flew backwards and hit him.

State and explain if his claim is valid.

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..... [2]

(ii) A labourer was tasked with pulling a cart. He reasoned that whatever he exerts on the cart, the cart will exert an equal and opposite force on him. The forces will cancel out and hence it is pointless for him to pull the cart as both he and the cart will not be able to move.

State and explain if his reasoning is correct.

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..... [2]

(iii) A man is stranded in the middle of a frozen lake with a heavy bag of gold. As there is no friction between him and the surface of the ice, he is unable to crawl back to shore.

State and explain the action that he can take to get back to shore.

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..... [2]

(c) A small sedan car makes a head-on collision with a large truck.

(i) State and explain if the force experienced by the car is different from that experienced by the truck.

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..... [1]

(ii) Both drivers are securely fastened to their vehicle seats. State and explain if the driver of the truck is likely to experience more severe injuries as compared to the driver of the car.

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